

Hassan Nawaz

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EDUCATION

Hong Kong University of Science and Technology Sep. 2026-July. 2030
Doctor of Philosophy (PhD) in Chemical and Biomolecular Engineering
Advisor: Hanyu Gao

Xiamen University Sep. 2024-July. 2026
Master of Science in Physical Chemistry
Advisor: Pavlo O. Dral
GPA: 3.9/4 (*top of class*)
Thesis: Exploring Automation and Acceleration in Computational Chemistry with Artificial Intelligence and Machine Learning Potentials.

University of Agriculture Faisalabad Sep. 2020-Jun. 2024
Bachelor of Science in Analytical Chemistry
Advisor: Raja Adil Sarfraz **GPA:** 3.71/4 (*in top 5*) **Final year report:** Revolutionizing quantum and computational chemistry using machine learning.

AWARDS

HKUST RedBird PhD Recruitment Award 2026-27 April. 2026
Postgraduate Scholarship at HKUST Feb. 2026
Chinese Government Scholarship Sep. 2024
Ehsas Undergraduate Merit Base Scholarship Sep. 2021
Higher Secondary Education Bright Student Scholarship April. 2018

PUBLICATIONS

Hu, J., Nawaz, H., Hou, Y. F., Chi, L., Tong, X. Y., Rui, Y., ... & Dral, P. O. (2026). Aitomia: An Agentic Framework for AI-Driven Atomistic and Quantum Chemical Simulations. *Journal of Chemical Theory and Computation*. (Hu, J., Nawaz, H., Hou, Y.F. are Co-1st authors)

RESEARCH EXPERIENCES

AI-driven Autonomous Computational Chemistry
(Got grant for this project from Natural Science Foundation of China (NSFC)) Sep. 2024-Present

- Fine-tuned domain-specific LLMs using LoRA and built RAG-based knowledge systems for enhanced reasoning in computational chemistry.
- Designed an autonomous simulation platform with multi-agent workflows (LangGraph) and A2A communication, integrating FastAPI for web deployment.
- Prepared and curated high-quality datasets for LLM training and optimized local deployment of customized models.

Development of Machine Learning Interatomic Potential beyond DFT accuracy
Collaboration with Nicolaus Copernicus University, Toruń, Poland with Ireneusz Grabowski and Szymon Śmiga (First-author manuscript will be submitted soon) Dec. 2024-Present

- Developed machine learning interatomic potentials aimed at surpassing standard DFT accuracy, including full workflow: data preparation, descriptor engineering (AEVs), and model training.
- Performed hyperparameter optimization and designed evaluation strategies such as learning curves and energy–force consistency checks.

Atomistic Simulations for Topological Materials Discovery

Collaboration with Anhui University with Arif Ullah

Sep. 2025-Present

- Conducting first-principles (DFT) calculations to identify and characterize non-trivial topological properties in quantum materials, including band topology, symmetry analysis, and topological invariant evaluation.
- Developing automation workflows and machine-learning-assisted screening tools to accelerate the discovery of new functional topological materials.

Ensemble learning for more accurate predictions

Oct. 2024-Jan. 2025

- Explored ensemble learning approaches inspired by group work showing that “the best DFT functional is the ensemble of functionals,” applying similar ideas to chemistry-focused machine learning tasks
- Designed and evaluated ensemble architectures—including linear regression and its variants—to combine outputs of multiple models trained on the ANI dataset for improved prediction accuracy.

Accurate calculation of heat of formation beyond harmonic oscillator approximations

Jan. 2024-Oct. 2024

- Investigated methods for improving thermochemical predictions—specifically heats of formation—by incorporating vibrational anharmonicity, and benchmarked multiple computational approaches against standard reference datasets

Conferences and Workshops

American Chemical Society (ACS) spring meeting at *Atlanta, Georgia, USA*

March 2026

- Will present poster on the title of How well AI Agents Perform as Expert Computational Chemists? Case studies with Aitomia.

Hands-on Workshop on Electronic-Structure Theory and AI for Materials Science, Shanghai University, China

Nov.2025

- Presented poster on the title of our Intelligent Assistant for AI-Driven Atomistic and Quantum Chemical Simulations.

International Conference of Computational Organic-synthesis Catalysis, Southeast University of Science and Technology, China

Oct.2025

- Presented poster on the title of ‘How well AI-agents perform as Expert Computational Chemist? Case studies with Aitomia.

Conference on Modeling and Design of Molecular Materials, Faculty of Chemistry, *Wroclaw University of Science and Technology, Poland*

Sep. 2025

- Presented poster on title of Aitomia: Your Intelligent Assistant for AI-Driven Atomistic and Quantum Chemical Simulations

The Fourth XACS Symposium on Chemical Bonds and AI Molecular Simulation (XACSW-2025) at Shaanxi Normal University Xi'an, China Aug. 2025

- Served as instructor for training students on AI-driven computational chemistry simulations

AI-Driven Discoveries: Machine Learning for Physical Sciences at Fudan University Shanghai April. 2025

- Participated in this conference

Annual Conference of Fujian Key Laboratory of Theoretical and Computational Chemistry, Chinese Academy of Sciences (UCAS), Fuzhou, China Jan. 2025

- Present poster as a representative of my whole research group on Updates on the MLATOM software package for atomistic simulations

International Workshop on Nanomaterials and Catalysis at Pakistan Institute of Engineering and Applied Sciences (PIEAS) May 2024

- Talk on Nanotechnology in improving cathodic properties for lithium-ion batteries using Olivine-based structured LiFePO₄

International Conference on Chemistry, Materials and Technology Tongji University of Agriculture Faisalabad (UAF) April. 2023

- Presented poster on the topic of Enhancing Olivine-Structured LiFePO₄ and won best poster prize.

International Symposium on Scientific Writing at UAF March. 2022

- Lead a team of 6 students under Dr. Asif Hanif to present science model based on biodiesel production and got 1st prize for the best science model in the conference.

WORK EXPERIENCE

Research Intern at Central Hi-Tech Lab, University of Agriculture Faisalabad Feb. 2023-Jul. 2023

(Advisors: Raja Adil Sarfraz, Raziya Nadeem, Fazil Shah)

- Development of novel methodologies for the identification of the adulterants in milk.
- Hands on Training on Scanning Electron Microscope (SEM), High-performance Liquid Chromatography (HPLC), Gas Chromatography (GC), Light Microscopy (LM), Atomic Absorption Spectroscopy (AAS), UV-visible spectroscopy, and Flame photometry (FP) etc.

SKILLS

Language: Urdu, Punjabi, Arabic, English (Undergraduate and master's all education is in English) **Computer:** Python, Linux, LaTeX